Mayank Kantharia

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Profile

Data Scientist with experience in machine learning, NLP, cloud-based systems, and end-to-end analytical solution development. Skilled at transforming complex datasets into actionable insights, building scalable ML services, and collaborating with stakeholders to deliver business-focused outcomes. Passionate about applying advanced analytics to solve real-world problems and support data-driven decision-making and operational improvements.

Key Skills

Machine Learning & Analytics: Supervised/unsupervised ML, NLP, LDA, sentiment analysis, embeddings, RAG pipelines, ranking models, model evaluation, statistical analysis, feature engineering, cross-validation

Programming & Frameworks: Python, R, FastAPI, Flask, Django, SQL, TypeScript (Next.js components)

AI & LLM Tools: OpenAI API (GPT-4o embeddings), vector search (pgvector), semantic re-ranking, skill-gap analysis, AI agent workflows

Cloud, APIs & Platforms: AWS (EC2, S3, DynamoDB, Lambda, API Gateway), REST API design & development, Supabase, Google Calendar API, Outlook Calendar API

DevOps & Engineering Practices: Docker, Git, GitHub, Agile/Scrum, CI/CD basics, logging & monitoring

Databases & Tools: MySQL, MongoDB, PostgreSQL, pgvector

Visualisation & BI: Tableau, Power BI, Matplotlib, Seaborn

Professional Strengths: Detail-oriented, analytical thinker, clear communicator, adaptable collaborator, well-organised

Internships

Path | Data Scientist Intern | Internship **July 2025 - Nov 2025**

- Built end-to-end RAG-based career guidance modules using GPT-4o embeddings and Supabase pgvector to generate personalized role recommendations and learning roadmaps.
- Developed FastAPI ML services for retrieval, semantic re-ranking, enrichment-pack generation, and skill-gap analysis, achieving 3 second end-to-end latency and 5 millisecond vector searches.
- Expanded internal datasets from 200 to 960 roles via automated web scraping and metadata enrichment, improving coverage and contextual accuracy.
- Integrated Google and Outlook Calendar APIs and contributed to Next.js and TypeScript components for roadmap visualization while collaborating in a 5-member Agile team.
- Implemented logging, monitoring, and evaluation scripts to support iterative experimentation and system reliability.

Projects

Cloud-Based Music Subscription Web Application | RMIT

AWS EC2, S3, DynamoDB, API Gateway, Lambda, Ubuntu, Apache2

- Developed a fully deployed cloud application hosted on an EC2 Ubuntu server using Apache2, accessible via public DNS.
- Designed and implemented DynamoDB schemas for login, music, and subscription data with optimized key structures.
- Built backend logic for login, registration, subscription management, and music queries using DynamoDB read/write operations.
- Automated image retrieval and upload by downloading artist images and storing them securely in S3.
- Integrated API Gateway and Lambda to process user registration, subscription updates, and data retrieval through REST APIs.
- Delivered a complete end-to-end cloud solution demonstrating scalable architecture and AWS best practices.

Russia-Ukraine War | Social Media & Network Analysis (Reddit + YouTube) | RMIT

Python, PRAW, YouTube API, NLP, LDA, VADER, NetworkX, Louvain

- Collected and processed 90,000 comments from Reddit and YouTube, performing large-scale cleaning and preprocessing.
- Built an NLP pipeline for sentiment analysis and LDA topic modeling to surface actionable insights.
- Constructed interaction graphs, applied Louvain community detection, and simulated information diffusion using the Independent Cascade model.
- Generated insights on sentiment polarization, topic diversity, and narrative spread across platforms.

NLP Clothing Review Classification & Web Application | RMIT

TensorFlow, FastText, TF-IDF, Logistic Regression, Flask

- Built end-to-end NLP pipelines for classifying product recommendations using text reviews.
- Implemented preprocessing: regex tokenization, stopword removal, rare-word filtering, vocabulary creation.
- Generated Bag-of-Words, TF-IDF, and embedding-based document vectors for model training.
- Compared multiple ML models using 5-fold cross-validation and evaluation metrics.
- Developed a Flask-based shopping website integrating the trained ML model for real-time recommendation prediction.
- Implemented keyword-based search, item browsing, and user review submission with dynamic ML inference.

Research Based Project on Deep Learning | Final Year Project | IIT Bombay

PyTorch, TensorFlow, Deep Learning, Attention Mechanisms

- Built a Few-Shot Learning model for hyperspectral image classification using high-dimensional spectral–spatial data, integrating Temperature Scaling and Attention mechanisms to improve predictive reliability.
- Performed preprocessing and feature extraction on curated HSI datasets to address spectral–spatial variability and enhance model generalization.
- Evaluated performance across multiple HSI benchmarks using cross-validation and ablation studies to quantify the impact of architectural components and tuning strategies.

Education

RMIT University, Melbourne (3.4/4 GPA) **2024-2025**

Master of Data Science

Mumbai University, Mumbai **2019-2023**

Bachelor of Computer Engineering (8.96/10 GPA)